

Full Length Article

A real-time infrared small target detector based on an illumination-aware degradation filtering network

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ABSTRACT

Infrared small-target detection remains challenging due to the extremely small size of targets, low target-background contrast, and severe interference from complex backgrounds and noise. To address these issues, this paper proposes an efficient real-time detection framework, termed IADF-YOLO, based on the YOLO paradigm. During training, a dual-backbone collaborative learning strategy is introduced to learn complementary feature representations. In addition, an illumination-aware degradation filtering network (IADF-Net) is designed to suppress illumination-induced degradation and noise-contaminated responses, while a content-aware cross-scale feature pyramid network (CACSPFN) is developed to enable efficient multi-scale feature fusion. Together, these components improve feature quality and multi-scale information utilization while preserving a lightweight architecture and real-time inference capability. Experiments on the ISA-TD and ISAD-6 datasets show that IADF-YOLO achieves a favorable balance among accuracy, model complexity, and inference speed. On ISA-TD, it achieves an mAP₅₀ of 0.936 and an mAP₅₀₋₉₅ of 0.740; on ISAD-6, it attains an mAP₅₀ of 0.949 with an F1 score of 0.92. With only 5.8M parameters and 11.9 GFLOPs, the proposed method consistently outperforms multiple state-of-the-art YOLO variants and other representative detectors. Further visualizations and qualitative comparisons show that IADF-YOLO maintains stable and reliable detection performance in practical scenarios involving complex backgrounds, long-range observation, and low-SNR conditions.

1. Introduction

In recent years, with the rapid development of deep learning and computer vision, object detection has made remarkable progress across a wide range of application domains. Among these, infrared small target detection has attracted increasing attention because of its critical importance in military surveillance, aerospace remote sensing, security monitoring, and disaster early warning systems (Zhao et al., 2022). Compared with visible-light targets, infrared small targets are typically characterized by extremely small size, low brightness, weak signal-to-noise ratio, and low contrast against complex backgrounds (Wu et al., 2021). Under interference from clouds, ground reflections, and infrared noise, the discriminative features of such targets are easily submerged, posing significant challenges to accurate detection (López-Alonso et al., 2002).

Early studies on infrared small target detection mainly relied on filtering techniques and mathematical models based on prior assumptions, such as background suppression, local contrast enhancement, and edge detection. Although these methods can achieve reasonable performance

under specific or controlled conditions, the complexity and variability of infrared imaging environments lead to highly uncertain background distributions. Consequently, rule-based approaches are often sensitive to parameter settings and show limited adaptability to scene variations and noise interference, making them insufficient to meet the practical requirements of detection accuracy, robustness, and real-time performance in complex scenarios (Zhao et al., 2019b).

In recent years, deep learning methods, especially convolutional neural networks (CNNs), have demonstrated significant advantages in infrared small target detection (Wang et al., 2019a; Ying et al., 2023). According to the detection pipeline, existing deep learning-based detectors can generally be divided into two-stage and single-stage approaches (Liu et al., 2016; Ren et al., 2017). Two-stage detectors first generate region proposals and then perform refined classification and regression, typically achieving high detection accuracy, particularly in complex backgrounds and low signal-to-noise conditions. However, their relatively high computational complexity limits their applicability in real-time systems. In contrast, single-stage detectors offer superior computational

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efficiency while maintaining competitive detection performance (Lin et al., 2017b), and have therefore gradually become the mainstream choice for infrared small target detection.

Within the single-stage detection framework, the YOLO (You Only Look Once) series has been widely adopted for infrared small target detection because of its end-to-end architecture and efficient inference capability (Bochkovskiy et al., 2020; Jocher, 2020; Jocher et al., 2023, 2025; Li et al., 2022; Redmon and Farhadi, 2018; Tian et al., 2025; Wang et al., 2024a, 2022, 2024b). Since the introduction of YOLOv1, the YOLO family has continuously evolved in both network architecture and detection mechanisms, achieving a favorable balance between detection accuracy and real-time performance. Early versions enhanced multi-scale detection capability through anchor-based designs and multi-scale prediction structures, while subsequent improvements in backbone networks, feature fusion strategies, and lightweight designs further boosted detection performance without sacrificing efficiency. More recently, the incorporation of anchor-free detection paradigms, lightweight network architectures, and attention mechanisms has further strengthened YOLO's ability to represent and localize small targets in complex backgrounds, expanding its potential for infrared small target detection tasks.

Nevertheless, existing YOLO-based methods still face several critical challenges in infrared small target detection scenarios. First, owing to uneven illumination, noise interference, and complex backgrounds in infrared imagery, small-target features are prone to severe degradation, while current models lack effective mechanisms to explicitly model and suppress such degraded features (Yang et al., 2023a). Second, traditional multi-scale feature fusion schemes struggle to preserve both fine-grained details and high-level semantic information for small targets across different scales (Zhu et al., 2025). Third, conventional IoU-based bounding-box regression metrics are highly sensitive to slight localization errors in small target scenarios, which limits further improvements in detection accuracy (Liu, 2024).

Based on the above analysis, this paper proposes a systematic enhancement framework for infrared small target detection built upon the YOLO architecture. Specifically, during training, an auxiliary backbone termed IADF-Net is introduced to explicitly model illumination information and suppress degraded features. Operating in parallel with the conventional YOLO backbone, IADF-Net extracts complementary feature representations to strengthen the discriminative capability of target features. In addition, CACSFPN is designed to perform content-adaptive feature reassembly and cross-scale interactive fusion on the multi-level features produced by the dual backbones, thereby improving multi-scale feature representation and fusion efficiency. To ensure real-time deployment, only the conventional YOLO backbone is retained during inference, while the IADF-Net branch is removed, resulting in significant improvements in detection accuracy and robustness with virtually no additional inference overhead. Extensive experimental results demonstrate that the proposed method substantially improves the accuracy and stability of infrared small target detection in complex backgrounds while maintaining efficient inference performance.

The remainder of this paper is organized as follows. Section 1 introduces the research background and objectives. Section 2 reviews related work and identifies the existing research gaps. Section 3 presents the proposed methodology and its implementation details. Section 4 reports the experimental results and provides an in-depth analysis. Finally, Section 4 concludes the paper.

1.1. Degradation suppression and discriminability enhancement for infrared small target detection

Under low signal-to-noise ratio and complex background conditions, infrared small target detection is often challenged by weak target responses, where target features are easily submerged by background clutter and noise (Wang et al., 2019a; Ying et al., 2023; Zhang et al., 2025). To address this issue, existing studies mainly focus on two represen-

tative directions, namely degradation suppression and discriminability enhancement (Cheng et al., 2024).

One line of research concentrates on background suppression and local contrast modeling, where saliency priors or contrast priors are introduced to improve the separability between targets and background. For example, PKNet (Yan et al., 2025) enhances detection capability in complex backgrounds by introducing a parallel interactive Kolmogorov-Arnold network. DNANet (Ren et al., 2021) and MLCL-Net (Yu et al., 2022) further strengthen small target representation through dense multi-level feature interaction and local contrast learning, respectively. However, most of these methods still rely on implicit feature learning and lack explicit modeling of degradation factors such as illumination variation and noise contamination, which limits their stability under complex imaging conditions.

Another line of research emphasizes explicit modeling and suppression of degradation mechanisms by addressing uneven illumination, noise interference, and artifacts either before detection or within the feature domain, thereby improving model robustness. For instance, Retinexformer handles non-uniform illumination through an illumination-guided mechanism, while FDI-Net (Chen et al., 2023) incorporates physical priors to enhance weak target responses. Nevertheless, these methods are primarily designed for general image enhancement or restoration tasks, and their objectives are not fully aligned with the discriminative representation requirements of infrared small target detection. As a result, their applicability to complex detection scenarios remains limited (Wang et al., 2025).

Overall, existing methods still struggle to simultaneously achieve effective degradation suppression and stable feature representation in complex infrared scenes. Motivated by this observation, this paper improves the detection framework from the perspective of explicit degradation modeling to enhance the stability of infrared small target representation under challenging backgrounds.

1.2. Multi-scale feature fusion and feature pyramid networks

In infrared small target detection, features at different scales exhibit strong complementarity. Shallow features preserve rich spatial details and positional cues, which are beneficial for precise localization of small targets, whereas deep features encode stronger semantic information that helps distinguish targets from the background. The Feature Pyramid Network (FPN) has become a classical framework for improving detection performance by enabling multi-scale information fusion through cross-layer connections (Lin et al., 2017a). However, due to the low contrast, severe noise interference, and complex backgrounds in infrared imagery, conventional FPNs are prone to information degradation and semantic imbalance during cross-layer fusion, thereby limiting the detection performance of small targets (Li et al., 2023).

To alleviate these limitations, various enhanced multi-scale fusion architectures have been proposed. For example, MLFPN improves multi-scale representation by combining a U-shaped topology with multi-level fusion modules (Yi et al., 2020). MLP-Net (Wang et al., 2024c) constructs a multilayer perceptron-based feature fusion network to improve inter-layer interaction and fusion effectiveness. In addition, methods such as Zoom Text Detector (Yang et al., 2023b), Reinforcement Shrink-Mask for Text Detection (Yang et al., 2023c), and CM-Net (Yang et al., 2022), although primarily developed for text detection, provide useful insights for this work through their designs for multi-scale feature fusion and complex background modeling. Nevertheless, these methods still have room for further improvement in handling the pronounced degradation characteristics, strong background interference, and real-time requirements of infrared scenarios (Wu et al., 2024).

Different from existing methods that mainly strengthen feature fusion by increasing connection density or constructing complex topological structures, the proposed CACSFPN focuses more on content-adaptive reassembly and cross-scale interaction. It aims to enhance semantic consistency across feature hierarchies while preserving the

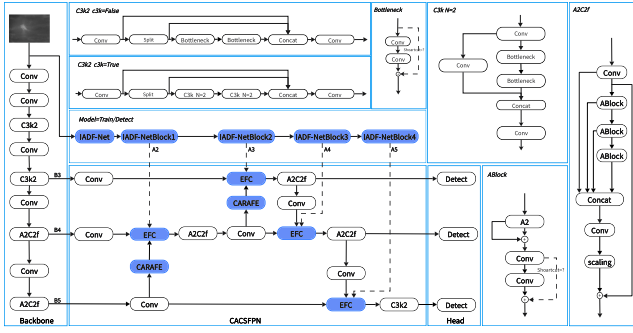


Fig. 1. Overview of the proposed dual-backbone framework for infrared small target detection.

fine-grained details required for small target detection as much as possible. This design is more compatible with the characteristics of infrared small targets, which typically exhibit weak responses, limited spatial support, and high susceptibility to background interference. Therefore, CACSPN provides a more effective multi-scale representation framework for small target detection in complex infrared scenes.

Overall, although existing methods have made progress in multi-scale feature fusion, the selectivity and effectiveness of the fusion process in complex infrared scenes still remain insufficient. To address this issue, this paper designs CACSFPN to further improve cross-level feature interaction and fusion quality, thereby enhancing the representation capability for infrared small targets.

In summary, although substantial progress has been made in degradation suppression and multi-scale feature fusion, explicit degradation modeling and effective feature integration mechanisms still require further improvement for infrared small target detection in complex scenes. Therefore, this paper improves the YOLO-based detection framework from the perspectives of degradation suppression and multi-scale feature fusion to enhance the accuracy, robustness, and efficiency of infrared small target detection under challenging backgrounds.

2. Methods

Building on the YOLOv12 detection framework (Tian et al., 2025), we propose *IADF-YOLO*, whose overall architecture is shown in Fig. 1. The model retains the A2C2f module of YOLOv12 as the basic feature extraction unit, in which ABlock and the A2 region-attention module enhance local feature interaction and representation. On this basis, IADF-Net, CACSFPN, and a dual-backbone collaborative learning strategy are introduced to improve degradation suppression, feature enhancement, and multi-scale fusion, respectively. These components jointly strengthen infrared small-target representation under complex backgrounds and low signal-to-noise conditions while preserving the lightweight design and real-time inference capability of YOLOv12.

The following sections present the structural designs and operating principles of these key modules in detail, and further analyze their effects on detection accuracy and computational efficiency.

2.1. Dual-backbone collaborative learning framework

Infrared small target detection remains challenging due to weak target responses and severe degradation effects. Infrared small targets usually occupy very limited spatial regions and contain only sparse structural cues. Meanwhile, infrared imagery is often degraded by non-uniform illumination, background fluctuations, and sensor noise, which jointly distort target responses and hinder stable feature learning. Under such conditions, relying on a single detection backbone to simultaneously perform degradation suppression and target discrimination often results in unstable feature representations and limited performance gains.

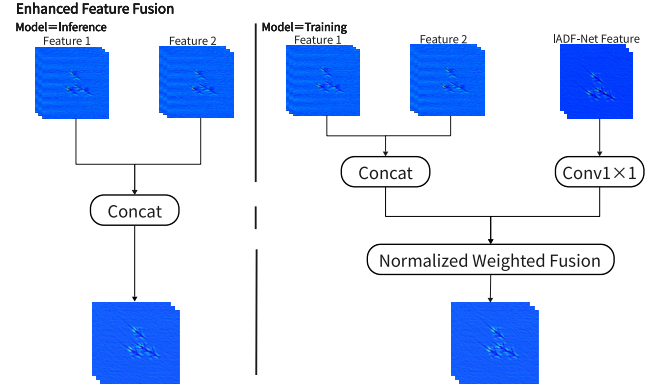


Fig. 2. Structure of the proposed EFC module.

To address this issue, we propose a dual-backbone collaborative learning framework, as illustrated in Fig. 1. Rather than simply increasing network capacity, the proposed framework decouples degradation modeling from target discrimination at the architectural level. Specifically, in addition to the conventional YOLO detection backbone, IADF-Net is introduced as an auxiliary backbone to explicitly model illumination variations and degradation factors in infrared imagery. In this way, the auxiliary branch is dedicated to generating stable degradation-aware features, while the main backbone focuses on localization and classification, thereby alleviating task interference between degradation suppression and target detection.

During training, the YOLO backbone and IADF-Net operate in parallel. Two corresponding feature maps from the main backbone are first concatenated to form a base fused representation, denoted by $\mathbf{F}_b$. Meanwhile, the auxiliary feature generated by IADF-Net is aligned through a 1×1 convolution to obtain $\mathbf{F}_f$. These two features are then fed into the Enhanced Feature Fusion (EFC) module, whose structure is shown in Fig. 2. To incorporate degradation-aware information without altering the output dimensionality, EFC adopts a normalized weighted fusion strategy:

$$\hat{w}_j = \frac{\phi(w_j)}{\sum_{k \in \{b, i\}} \phi(w_k) + \varepsilon}, \quad j \in \{b, i\}, \quad (1)$$

$$\mathbf{F}_{\text{out}} = \hat{w}_b \mathbf{F}_b + \hat{w}_i \mathbf{F}_i.$$

where w_j is a learnable weight, $\phi(\cdot)$ denotes a nonlinear activation function implemented as Swish, and ε is a small constant for numerical stability.

During inference, only the concatenation path of the main backbone shown in Fig. 1 is retained, while the auxiliary IADF-Net branch is removed. Because EFC introduces the auxiliary feature through dimension-preserving weighted fusion rather than channel expansion, removing the auxiliary branch does not alter the feature dimensionality fed into CACSFPN or introduce weight mismatch in the subsequent convolutional layers. In this way, the proposed framework preserves the robustness gains brought by degradation-aware guidance during training while maintaining high inference efficiency.

Overall, the proposed dual-backbone collaborative learning framework introduces an auxiliary degradation-modeling branch during training and stabilizes cross-branch feature enhancement through EFC. As a result, it provides an effective solution for infrared small target detection that jointly balances accuracy, robustness, and real-time deployability.

2.2. Illumination-aware degradation filtering network

In infrared small target detection, targets typically exhibit extremely small spatial extent, low contrast, and weak structural cues, making them highly susceptible to background clutter and noise. Therefore, explicitly modeling the interplay among targets, background, and

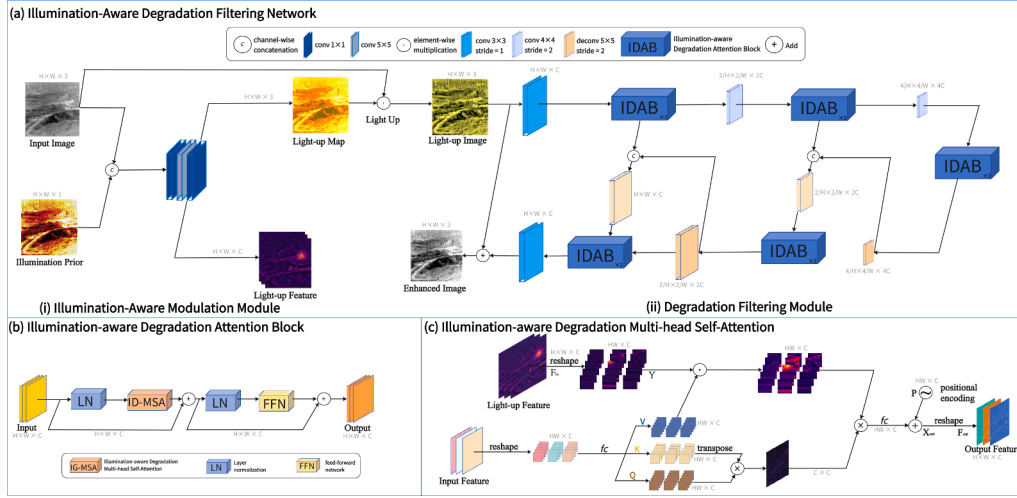


Fig. 3. (a) Our method adopts the proposed one-stage illumination-aware framework, which consists of an illumination modulation module (i) IAMM and a degradation filtering network (ii) IADF-Net. (b) The basic unit of IADF-Net is IDAB, composed of two LN layers, an ID-MSA module, and an FFN. (c) ID-MSA uses the illumination/modulation guidance extracted by IAMM to direct self-attention computation.

illumination is essential for improving target-background separability and detection robustness. The overall architecture of the proposed IADF-Net is illustrated in Fig. 3.

From the perspective of infrared imaging, an infrared image can be viewed as a scene structure modulated by illumination conditions. Let the input infrared image be denoted by $\mathbf{X} \in \mathbb{R}^{H \times W \times 3}$, which can be formulated as the element-wise coupling of structural and illumination components:

$$\mathbf{X} = \mathbf{S} \odot \mathbf{T}, \quad (2)$$

where $\mathbf{S} \in \mathbb{R}^{H \times W \times 3}$ denotes the structural response containing potential small targets and background information, $\mathbf{T} \in \mathbb{R}^{H \times W}$ represents the illumination (or radiation intensity) distribution, and $\odot$ denotes element-wise multiplication.

The above formulation assumes an ideal degradation-free imaging process. In practical infrared scenarios, however, this assumption is often violated by sensor noise, environmental interference, and illumination instability. To explicitly characterize such degradation effects, perturbation terms are introduced into both the structural and illumination components:

$$\mathbf{X} = (\mathbf{S} + \Delta\mathbf{S}) \odot (\mathbf{T} + \Delta\mathbf{T}), \quad (3)$$

where $\Delta\mathbf{S}$ denotes structure-related degradation caused by noise and background clutter, and $\Delta\mathbf{T}$ represents illumination-related degradation induced by uneven radiation or estimation errors.

Direct illumination estimation followed by element-wise division may cause numerical instability, particularly in low-radiation regions. To avoid this issue, an illumination modulation strategy is adopted. Specifically, an adaptive illumination modulation map $\mathbf{M} \in \mathbb{R}^{H \times W \times 3}$ is introduced to adjust image brightness in a multiplicative manner:

$$\mathbf{X}_{\text{mod}} = \mathbf{X} \odot \mathbf{M}. \quad (4)$$

By incorporating the above degradation formulation, the modulated image can be rewritten as

$$\mathbf{X}_{\text{mod}} = \mathbf{S} + \mathbf{D}, \quad (5)$$

where $\mathbf{D} \in \mathbb{R}^{H \times W \times 3}$ denotes the aggregated degradation term amplified during illumination modulation.

Based on the above analysis, we establish a unified two-stage framework composed of an illumination-aware modulation module and a degradation filtering module. Specifically, the illumination-aware modulation module $\mathcal{E}(\cdot)$ takes the input image $\mathbf{X}$ and an illumination prior $\mathbf{T}_p$ as input, and produces the modulated image $\mathbf{X}_{\text{mod}}$ together with an

illumination-guidance feature $\mathbf{F}_{\text{mod}}$. The degradation filtering module $\mathcal{R}(\cdot)$ then uses $\mathbf{F}_{\text{mod}}$ to suppress non-target degraded responses in $\mathbf{X}_{\text{mod}}$, yielding a cleaner representation $\mathbf{X}_{\text{clean}}$ for subsequent localization and classification:

$$(\mathbf{X}_{\text{mod}}, \mathbf{F}_{\text{mod}}) = \mathcal{E}(\mathbf{X}, \mathbf{T}_p), \quad \mathbf{X}_{\text{clean}} = \mathcal{R}(\mathbf{X}_{\text{mod}}, \mathbf{F}_{\text{mod}}). \quad (6)$$

The illumination prior $\mathbf{T}_p \in \mathbb{R}^{H \times W}$ is obtained by channel-wise averaging:

$$\mathbf{T}_p = \text{Mean}_c(\mathbf{X}). \quad (7)$$

It is worth noting that the proposed Illumination-Aware Degradation Filtering Network (IADF-Net) is a concrete realization of the degradation filtering module $\mathcal{R}(\cdot)$ within the unified framework, and the two correspond one-to-one in both formulation and functionality. Rather than serving as a simple enhancement of a conventional detection backbone, IADF-Net is explicitly derived from the aforementioned infrared imaging mechanism and degradation model. By introducing illumination information as explicit guidance into feature interaction, IADF-Net enables the network to distinguish non-target degraded responses caused by illumination non-uniformity and noise from stable structural responses associated with true infrared small targets, thereby achieving targeted degradation suppression and discriminative feature enhancement in the feature domain.

Although IADF-Net is inspired by illumination-aware and Retinex-based modeling, its design objective and implementation paradigm differ fundamentally from those of existing image enhancement methods. Models such as RetinexNet (Wei et al., 2018) and Retinexformer (Cai et al., 2023) are primarily developed for generic image enhancement or restoration, with emphasis on image decomposition, illumination correction, and perceptual quality improvement. In contrast, IADF-Net is not intended to generate visually enhanced images. Instead, it is specifically designed for infrared small target detection, where illumination-guided information is explicitly incorporated into feature learning to suppress non-target degraded responses and enhance stable target-related structural representations. Moreover, IADF-Net is not used as an independent preprocessing module. Rather, it is embedded into the detection framework as an auxiliary backbone and jointly optimized with the main detection backbone under detection-oriented supervision. This task-driven design makes it better suited to infrared small target detection, where weak-target preservation, background suppression, and real-time performance must all be considered.

Most existing infrared small target detection methods primarily rely on convolutional neural networks, which are effective at modeling local

spatial patterns but remain limited in capturing long-range spatial dependencies. Although attention mechanisms can improve global feature interaction, conventional global self-attention usually incurs prohibitive computational and memory costs on high-resolution feature maps, making it unsuitable for real-time infrared small target detection.

To address these limitations, we introduce the Illumination-Aware Degradation Filtering Network (IADF-Net) as the degradation filtering module $\mathcal{R}(\cdot)$ within the proposed unified framework. As shown in Fig. 3, IADF-Net adopts a hierarchical encoder-decoder architecture that enables cross-level feature interaction while maintaining computational efficiency. In addition, illumination-aware degradation attention is incorporated at key feature transformation stages to strengthen global modeling under complex imaging conditions. By explicitly injecting illumination or intensity distributions as conditional guidance into feature learning, IADF-Net effectively suppresses background clutter and noise-induced degradation while enhancing stable target-related structural responses. As a result, it provides cleaner and more robust representations for subsequent multi-scale feature fusion and detection-head regression.

Network architecture. IADF-Net is built upon a hierarchical encoder-decoder feature interaction framework, enabling progressive abstraction and cross-level fusion. Its input is the modulated image $\mathbf{X}_{\text{mod}}$.

In the encoding stage, the input is first processed by a 3×3 convolution, followed by an Illumination-Aware Degradation Attention Block (IDAB), and then by a 4×4 convolution with stride 2 for spatial downsampling. Subsequently, two IDABs and another 4×4 strided convolution are applied to obtain multi-level feature representations:

$$\mathbf{F}_i \in \mathbb{R}^{\frac{H}{2^i} \times \frac{W}{2^i} \times C}, \quad i = 0, 1, 2. \quad (8)$$

At the deepest level, two additional IDABs are employed to further strengthen global contextual modeling.

In the decoding stage, spatial resolution is progressively restored using 2×2 transposed convolutions and skip connections to alleviate information loss caused by downsampling. The network outputs a residual term

$$\begin{aligned} \mathbf{X}_{\text{res}} &\in \mathbb{R}^{H \times W \times 3}, \\ \mathbf{X}_{\text{clean}} &= \mathbf{X}_{\text{mod}} + \mathbf{X}_{\text{res}}. \end{aligned} \quad (9)$$

Here, $\mathbf{X}_{\text{clean}}$ denotes the filtered and enhanced output, which is subsequently fed to the detection head for bounding box regression and category prediction. **Illumination-aware degradation attention.** The core component of IADF-Net is the Illumination-Aware Degradation Multi-Head Self-Attention (ID-MSA). Given an input feature map $\mathbf{F}_{\text{in}} \in \mathbb{R}^{H \times W \times C}$, it is reshaped into token representations:

$$\mathbf{X} \in \mathbb{R}^{HW \times C}. \quad (10)$$

The tokens are then split into k attention heads, where $k = 4$ in all experiments:

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k], \quad \mathbf{X}_i \in \mathbb{R}^{HW \times d_k}, \quad d_k = \frac{C}{k}. \quad (11)$$

For each head, the query, key, and value matrices are computed as

$$\mathbf{Q}_i = \mathbf{X}_i \mathbf{W}_{Q_i}^T, \quad (12)$$

$$\mathbf{K}_i = \mathbf{X}_i \mathbf{W}_{K_i}^T, \quad (13)$$

$$\mathbf{V}_i = \mathbf{X}_i \mathbf{W}_{V_i}^T. \quad (14)$$

The illumination-guidance feature $\mathbf{F}_{\text{mod}}$ produced by $\mathcal{E}(\cdot)$ is reshaped and split in the same manner:

$$\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_k]. \quad (15)$$

Since IADF-Net operates at multiple scales, $\mathbf{F}_{\text{mod}}$ is resized, e.g., via stride-2 convolutions or equivalent scale transformations, to match the spatial resolution of the current feature level before forming $\mathbf{Y}_i$.

The attention operation is defined as

$$\text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i, \mathbf{Y}_i) = (\mathbf{Y}_i \odot \mathbf{V}_i) \text{softmax}\left(\frac{\mathbf{K}_i^T \mathbf{Q}_i}{\alpha_i}\right), \quad (16)$$

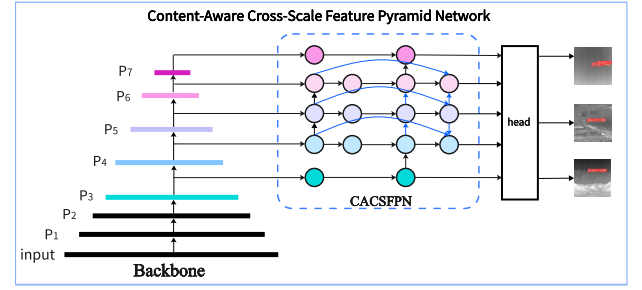


Fig. 4. CACSFPN network architecture diagram.

where α_i is a learnable scaling parameter.

Computational complexity. The computational cost of ID-MSA is dominated by matrix multiplication and can be expressed as

$$O(\text{ID-MSA}) = \frac{2HWC^2}{k}. \quad (17)$$

In contrast, the complexity of conventional global multi-head self-attention is

$$O(\text{G-MSA}) = 2(HW)^2C. \quad (18)$$

Therefore, ID-MSA has linear complexity with respect to spatial resolution, enabling efficient global modeling across multiple layers for infrared small target detection.

2.3. Content-aware cross-scale feature pyramid network

In infrared small target detection, multi-scale features exhibit strong complementarity. Shallow features preserve rich spatial details and positional cues, which are beneficial for precise localization of small targets, whereas deep features provide stronger semantic information for target-background discrimination. However, due to the low contrast, heavy noise, and complex backgrounds commonly found in infrared imagery, conventional feature pyramid networks often suffer from feature redundancy, semantic inconsistency, and information degradation during cross-scale fusion. Although IADF-Net can effectively suppress degradation and improve feature stability during training, these high-quality features cannot be fully exploited without an efficient multi-scale fusion mechanism. The overall architecture of the proposed CACSFPN is illustrated in Fig. 4.

To address this issue, we propose the Content-Aware Cross-Scale Feature Pyramid Network (CACSFPN) to improve multi-scale feature representation and fusion efficiency. During training, CACSFPN simultaneously receives multi-level features from the YOLO backbone and the auxiliary IADF-Net branch, and establishes a one-to-one correspondence between features at the same semantic level. Let the features from the YOLO backbone be denoted by $\{B_3, B_4, B_5\}$ and those from IADF-Net by $\{A_3, A_4, A_5\}$, where the subscripts denote different feature levels corresponding to feature maps downsampled by factors of 8, 16, and 32 relative to the input image. In other words, B_3 corresponds to A_3 , B_4 corresponds to A_4 , and B_5 corresponds to A_5 , and each paired feature set is used as input to CACSFPN.

Before cross-branch and cross-scale fusion, feature alignment is first performed. For each pair of corresponding features, a 1×1 convolution is applied to project them into a unified channel dimension, which can be written as

$$\tilde{B}_s = \text{Conv}_{1 \times 1}(B_s), \quad \tilde{A}_s = \text{Conv}_{1 \times 1}(A_s), \quad s \in \{3, 4, 5\}. \quad (19)$$

If the features from different sources have inconsistent spatial resolutions, upsampling or downsampling is further applied to align them to the same scale, thereby enabling subsequent element-wise fusion. After alignment, the features at each corresponding scale are fused to obtain a unified representation F_s . During inference, since IADF-Net is removed, CACSFPN takes only the multi-level features from the YOLO backbone

as input. Its hierarchical structure remains unchanged from training, and thus no dimensional mismatch is introduced into the subsequent modules.

The core of CACSFPN consists of CARAFE (Wang et al., 2019b) and the Cross-Scale Feature Fusion Module (CSFM). CARAFE mainly performs content-aware feature reassembly and upsampling. In our implementation, the local neighborhood size of the CARAFE content encoder is set to $k_{\text{enc}} = 3$. For an output position y , CARAFE extracts a local neighborhood $\mathcal{N}_{y, k_{\text{enc}}}$ of size $k_{\text{enc}} \times k_{\text{enc}}$ from the input feature map and generates position-specific reassembly weights through a content encoder:

$$W_{\Delta y} = \text{Softmax}(\psi(\mathbf{X}_{y+\Delta})), \quad \sum_{\Delta} W_{\Delta y} = 1, \quad (20)$$

where $\psi(\cdot)$ denotes a linear mapping implemented by a 1×1 convolution, and Δ represents the spatial offsets within the local neighborhood. The upsampled feature can then be expressed as

$$\mathbf{Y}_y = \sum_{\Delta \in \mathcal{N}_{y, k_{\text{enc}}}} W_{\Delta y} \cdot \mathbf{X}_{y+\Delta}. \quad (21)$$

Through this process, CARAFE better preserves local structural details during upsampling while suppressing the amplification of background noise.

After local reassembly, CSFM further performs cross-scale semantic interaction. Let the input features be denoted by $\{F_3, F_4, F_5\}$. CSFM first unifies the channel dimensions across different scales through 1×1 convolutions:

$$F'_s = \text{Conv}_{1 \times 1}(F_s). \quad (22)$$

The features from different scales are then resized to a target scale and fused by element-wise summation. For a target scale t , the fused feature can be written as

$$G_t = \sum_{s \in \Omega_t} \text{Resize}_t(F'_s), \quad (23)$$

where Ω_t denotes the set of feature levels involved in fusion, and $\text{Resize}_t(\cdot)$ denotes the spatial transformation that adjusts a feature map to the target scale t . In this way, fine-grained positional cues from high-resolution features and high-level semantic information from low-resolution features are effectively integrated into a unified representation space.

Overall, CACSFPN first integrates the dual-backbone inputs through explicit hierarchical mapping and feature alignment, then employs CARAFE for content-aware local reassembly, and finally uses CSFM to achieve cross-scale semantic fusion. This design enables the high-quality, degradation-suppressed features provided by IADF-Net during training to be exploited more effectively, thereby significantly improving the separability and localizability of infrared small targets in complex backgrounds. During inference, CACSFPN relies only on the outputs of the conventional detection backbone, thereby improving detection accuracy and robustness while maintaining low computational overhead.

3. Experiments

3.1. Dataset

The experiments in this study are conducted on two datasets. The first dataset is the Infrared Small Aircraft Target Detection (ISA-TD) dataset (Hui et al., 2019), which is designed for the detection and tracking of low-altitude small aircraft targets. It includes multiple target categories, such as helicopters, drones, small passenger aircraft, fighter jets, reconnaissance aircraft, and gliders. In total, ISA-TD contains 16,177 frames and 16,944 annotated targets. The distribution of annotated bounding boxes across categories is approximately as follows: 2,100 for helicopters, 2500 for drones, 1800 for small passenger aircraft, 2200 for fighter jets, 1600 for reconnaissance aircraft, and 1200 for gliders. The

Table 1
Experimental settings.

Experiment setup	Version
Hardware accelerator	NVIDIA GeForce RTX 4090
PyTorch	1.12.1
CUDA	11.3
Python	3.9.7
Input image size	640 × 640 pixels
Batch size	16
Learning rate	0.001
Optimizer	Adam
Weight decay	0.0005
Epochs	100

training, validation, and test splits are carefully designed to ensure that the model generalizes well across diverse scenarios.

The second dataset is a self-constructed infrared small aircraft detection dataset, termed ISAD-6. ISAD-6 consists of 6000 precisely annotated infrared images covering six representative categories of small aircraft targets, including helicopters, drones, gliders, small passenger aircraft, fighter jets, and reconnaissance aircraft. The dataset was collected in a variety of environments, such as urban, mountainous, and airport scenes, and includes aircraft images captured at different altitudes, speeds, and background conditions. To ensure diversity and representativeness, the dataset also includes images acquired under complex backgrounds and low signal-to-noise conditions.

All annotations were manually completed by professional annotators using two widely adopted image annotation tools, namely LabelImg and VIA (VGG Image Annotator). These tools were used to annotate bounding boxes and polygons for each target, thereby ensuring precise target localization in each image. The distribution of annotated bounding boxes in ISAD-6 is approximately as follows: 1,200 for helicopters, 1500 for drones, 1000 for small passenger aircraft, 1200 for fighter jets, 900 for reconnaissance aircraft, and 700 for gliders. These annotations provide accurate target location information for each image, thereby ensuring the applicability and robustness of the dataset across different operational scenarios.

All datasets are divided into training (70%), validation (15%), and test (15%) sets to ensure data diversity and improve the generalization capability of the model. Meanwhile, to ensure a fair comparison, all competing models and the proposed model are trained under the same training parameters and experimental settings listed in Table 1, including key hyperparameters such as input size, batch size, learning rate, number of training epochs, and optimizer. Except for differences in network architecture, all other training procedures and experimental conditions are kept identical, thereby ensuring the objectivity and comparability of the performance evaluation across different methods.

3.2. Feature visualization analysis

To provide deeper insight into the feature representation behavior of the proposed IADF-YOLO detector, we visualize and compare the multi-level feature responses produced by different methods, as shown in Fig. 5. The comparison includes RT-DETRv2 (Wang et al., 2019b), LCAE-Net (Yu et al., 2022), YOLOv12 (Tian et al., 2025), FDI-Net (Chen et al., 2023), and the proposed IADF-YOLO, thereby illustrating how different designs capture infrared small targets under complex background conditions.

From shallow to deep layers, most methods exhibit strong responses to background textures, structural clutter, and noise in the early stages, where target cues remain weak and are not well separated from the background. As the network depth increases, more discriminative semantic information gradually emerges, leading to increasingly concentrated responses around target regions. However, due to the inherent characteristics of infrared small targets, including low contrast, limited spatial extent, and weak structural signatures, several methods still

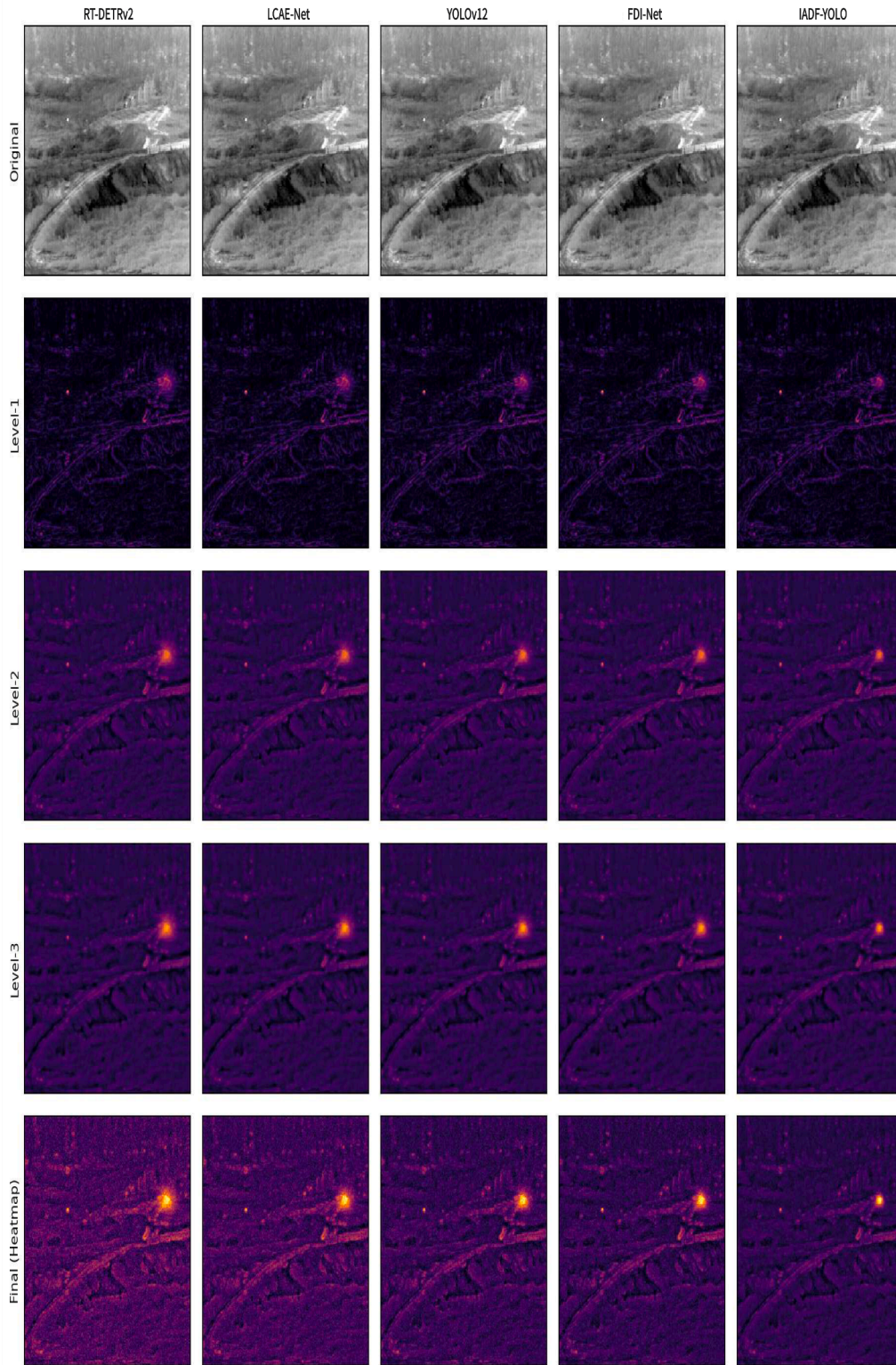


Fig. 5. Visualization results of feature maps at different levels for various models.

exhibit scattered activations or residual background interference in the intermediate and deep layers.

In contrast, IADF-YOLO demonstrates a more stable and coherent feature evolution process across different network levels. Benefiting from the illumination-aware degradation filtering mechanism, non-target responses caused by uneven illumination and noise are

effectively suppressed during feature extraction, allowing the network to progressively focus its representational capacity on true target regions. In deeper layers and in the final aggregated heatmaps, IADF-YOLO produces stronger and more compact activations at target locations, while substantially reducing background responses, indicating superior target-background separability.

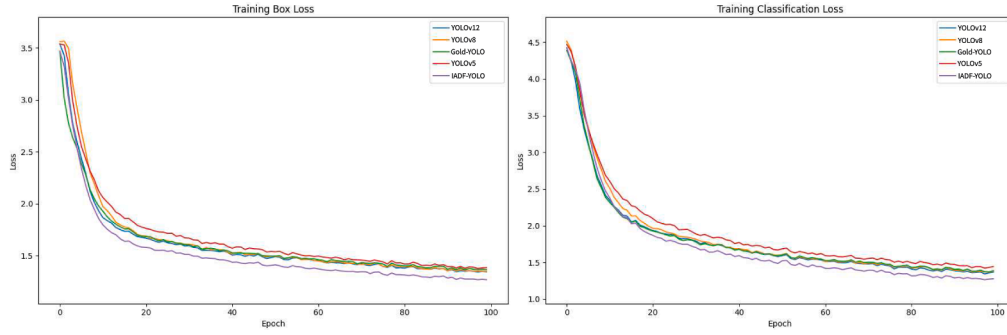


Fig. 6. Training loss curves of different models on the ISAD-6 dataset.

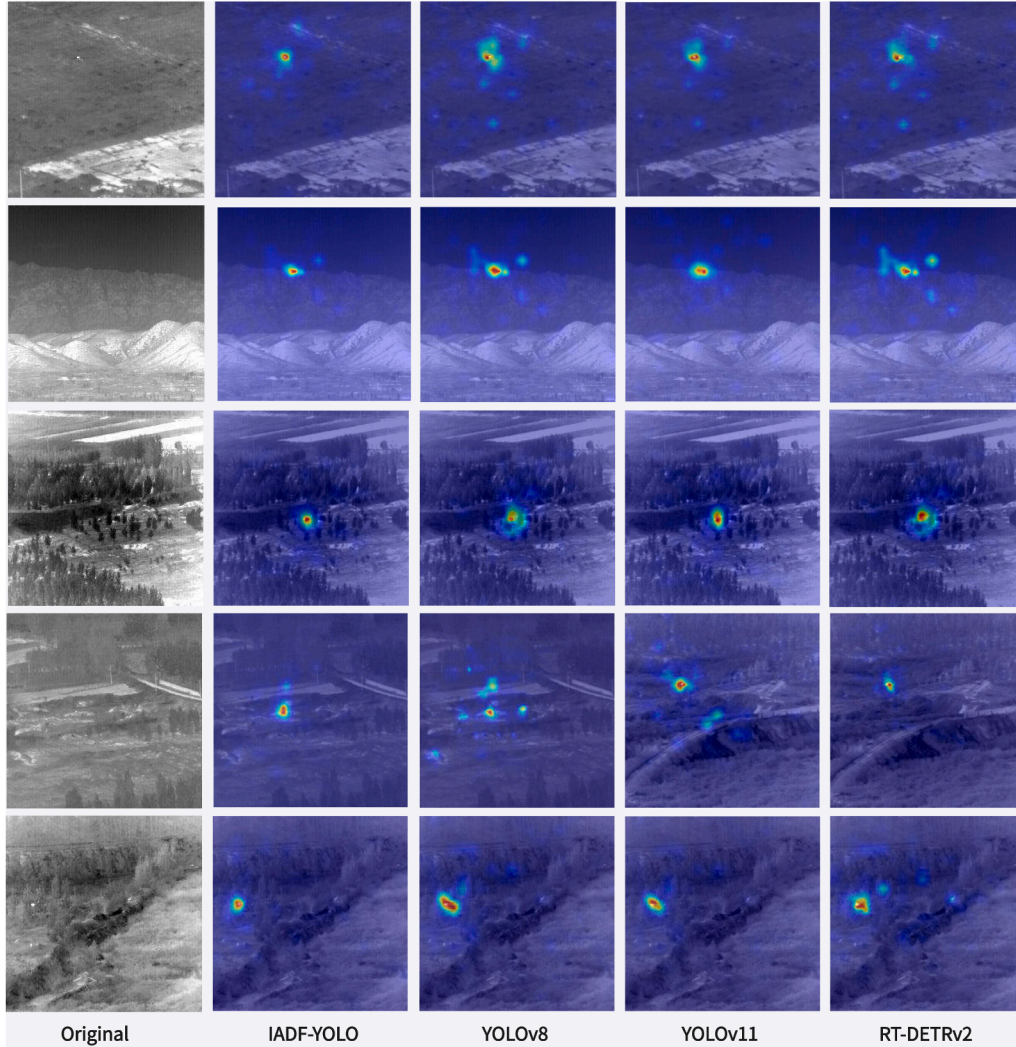


Fig. 7. CAM visualization results.

These visualizations demonstrate that IADF-YOLO is able to preserve fine-grained details in shallow layers while enhancing semantic consistency in deeper layers, thereby providing more discriminative and interpretable feature representations for robust infrared small target detection in complex scenes.

3.3. Comparative experiment

This section compares the proposed IADF-YOLO with multiple real-time detectors from the YOLO family and other representative object

detection models (Bochkovskiy et al., 2020; Jocher, 2020; Jocher et al., 2023, 2025; Li et al., 2022; Redmon and Farhadi, 2018; Tian et al., 2025; Wang et al., 2024a, 2022, 2024b). Experiments are conducted on two infrared small target datasets, ISA-TD and ISAD-6, using evaluation metrics including the number of parameters, computational complexity, mAP, FPS, and F1 score.

The results indicate that IADF-YOLO achieves the best or near-best overall performance on both datasets. On the ISA-TD dataset, as shown in Table 2, IADF-YOLO attains an mAP_{50} of 0.936 and an mAP_{50-95} of 0.740, outperforming YOLOv12-S, YOLOv11-S, and RT-DETRv2-L by

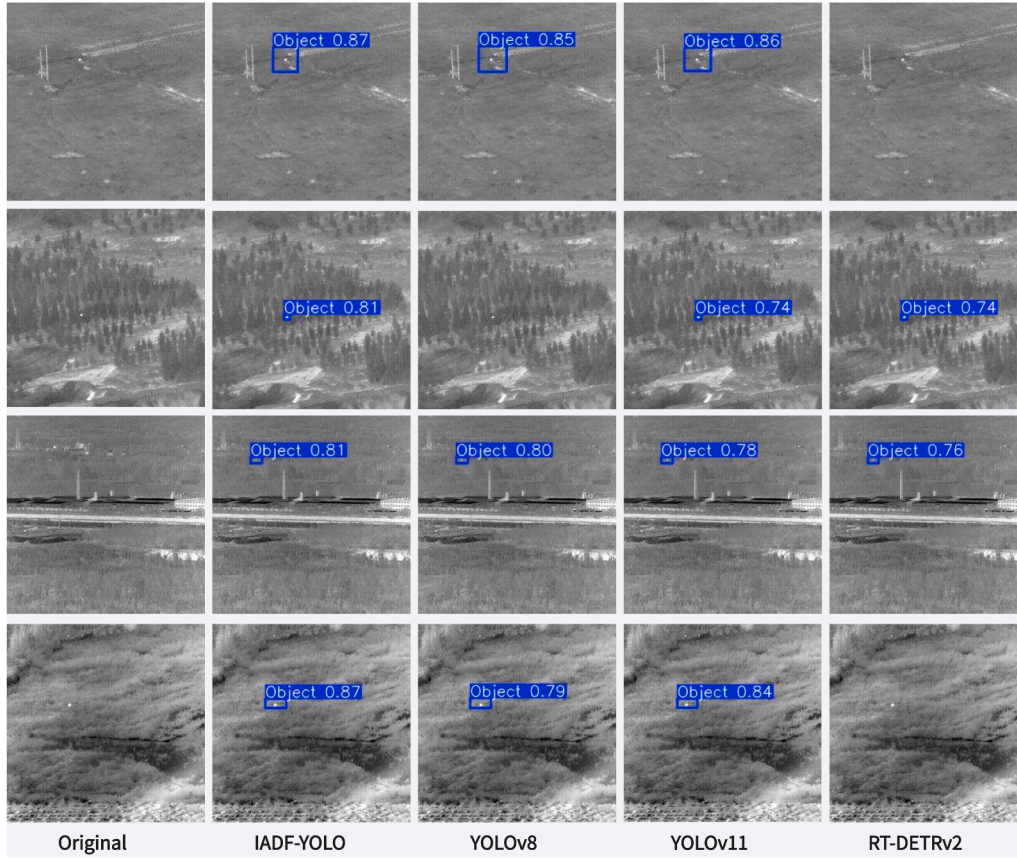


Fig. 8. Visualize the prediction results of different models.

Table 2

Performance comparison of different models on the ISA-TD dataset.

Model	mAP ₅₀	mAP ₅₀₋₉₅	FPS	Params (M)	GFLOPs	F1
SSD (Liu et al., 2016)	0.853	0.639	22.1	25.5	75.7	0.76
Faster R-CNN (Ren et al., 2017)	0.873	0.654	34.5	129.0	258.6	0.80
DETR (Carion et al., 2020)	0.886	0.647	24.1	159.0	306.0	0.81
DANet (Zhao et al., 2019a)	0.858	0.643	26.7	59.0	44.0	0.77
LCAE-Net (Yu et al., 2022)	0.916	0.659	39.6	35.6	63.1	0.75
FDI-Net (Chen et al., 2023)	0.920	0.671	41.3	39.6	59.1	0.79
DCCS-Det (Li et al., 2025b)	0.928	0.731	46.2	7.12	8.02	0.86
ILNet (Li et al., 2025a)	0.924	0.726	45.2	6.91	7.42	0.85
RT-DETRv1-L (Zhao et al., 2024)	0.930	0.732	41.7	32.0	110.0	0.86
RT-DETRv2-L (Lv et al., 2024)	0.932	0.736	44.5	31.0	109.0	0.89
YOLOv5-S (Jocher, 2020)	0.911	0.701	52.0	7.2	16.5	0.84
YOLOv6-S (Li et al., 2022)	0.903	0.661	59.2	16.3	44.2	0.86
YOLOv7-S (Wang et al., 2022)	0.921	0.721	45.1	36.0	104.0	0.84
YOLOv8-S (Jocher et al., 2023)	0.915	0.715	58.3	11.1	28.6	0.85
YOLOv9-S (Wang et al., 2024b)	0.919	0.715	55.1	8.2	22.8	0.85
YOLOv10-S (Wang et al., 2024a)	0.918	0.717	59.7	8.1	24.9	0.86
YOLOv11-S (Jocher et al., 2025)	0.925	0.720	60.3	8.8	21.0	0.87
Gold-YOLO-S (Wang et al., 2023)	0.912	0.723	45.1	21.5	46.0	0.86
YOLOv12-S (Tian et al., 2025)	0.927	0.721	58.6	9.2	21.7	0.87
IADF-YOLO (Ours)	0.936	0.740	61.1	5.8	11.9	0.89

Table 3

Performance comparison of different models on the ISAD-6 dataset.

Model	mAP ₅₀	mAP ₅₀₋₉₅	FPS	Params (M)	GFLOPs	F1
SSD (Liu et al., 2016)	0.883	0.521	22.1	25.5	75.7	0.79
Faster R-CNN (Ren et al., 2017)	0.892	0.531	34.5	129.0	258.6	0.81
DETR (Hui et al., 2019)	0.901	0.539	24.1	159.0	306.0	0.82
DANet (Carion et al., 2020)	0.885	0.523	26.7	59.0	44.0	0.79
LCAE-Net (Yu et al., 2022)	0.923	0.661	39.6	35.6	63.1	0.76
FDI-Net (Chen et al., 2023)	0.929	0.679	41.3	39.6	59.1	0.78
DCCS-Det (Li et al., 2025b)	0.937	0.605	46.2	7.12	8.02	0.86
ILNet (Li et al., 2025a)	0.934	0.600	45.2	6.91	7.42	0.85
RT-DETRv1-L (Zhao et al., 2024)	0.940	0.615	41.7	32.0	110.0	0.89
RT-DETRv2-L (Lv et al., 2024)	0.943	0.616	44.5	31.0	109.0	0.90
YOLOv5-S (Jocher, 2020)	0.919	0.575	52.0	7.2	16.5	0.86
YOLOv6-S (Li et al., 2022)	0.908	0.586	59.2	16.3	44.2	0.81
YOLOv7-S (Wang et al., 2022)	0.928	0.594	45.1	36.0	104.0	0.85
YOLOv8-S (Jocher et al., 2023)	0.933	0.613	58.3	11.1	28.6	0.88
YOLOv9-S (Wang et al., 2024b)	0.935	0.615	55.1	8.2	22.8	0.89
YOLOv10-S (Wang et al., 2024a)	0.939	0.618	59.7	8.1	24.9	0.90
YOLOv11-S (Jocher et al., 2025)	0.940	0.615	60.3	8.8	21.0	0.91
Gold-YOLO-S (Wang et al., 2023)	0.936	0.600	45.1	21.5	46.0	0.89
YOLOv12-S (Tian et al., 2025)	0.941	0.616	59.2	9.2	21.7	0.91
IADF-YOLO (Ours)	0.949	0.624	61.1	5.8	11.9	0.92

approximately 2.2%, 1.1%, and 0.4% in mAP₅₀, respectively. At the same time, it maintains the smallest parameter count (5.8M), low computational cost (11.9 GFLOPs), and the highest inference speed (61.1 FPS), demonstrating an excellent balance between accuracy and efficiency.

Compared with recent infrared-oriented detectors such as DCCS-Det and ILNet, IADF-YOLO also shows clearer overall advantages. Although both methods achieve competitive results with relatively lightweight designs, their mAP₅₀ scores remain lower than that of IADF-YOLO. More-

over, IADF-YOLO outperforms them in mAP₅₀₋₉₅, F1 score, and inference speed. While DCCS-Det and ILNet require slightly fewer GFLOPs, IADF-YOLO uses fewer parameters and achieves better overall accuracy, indicating more effective utilization of model capacity for infrared small target detection.

These results demonstrate that the proposed method achieves a more favorable trade-off among detection accuracy, robustness, model compactness, and real-time deployment capability, further confirming the effectiveness and efficiency of IADF-YOLO in complex infrared scenes.

Table 4

Ablation study of IADF-Net, CACSPFN, and TwoBackbone.

IADF-Net	CACSPFN	TwoBackbone	Params (M)	GFLOPs	mAP ₅₀	mAP ₅₀₋₉₅	FPS
×	×	×	9.2	21.7	0.941	0.616	59.2
✓	×	×	9.2	21.7	0.942	0.617	59.2
×	✓	×	4.4	10.4	0.942	0.619	62.5
×	×	✓	12.1	25.6	0.947	0.622	57.3
✓	✓	×	4.4	10.4	0.943	0.620	62.5
×	✓	✓	5.8	11.9	0.949	0.624	61.1

On the ISAD-6 dataset, IADF-YOLO again achieves the best overall performance. As shown in Table 3, it attains an mAP₅₀ of 0.949 and an mAP₅₀₋₉₅ of 0.624. Compared with YOLOv8-S and YOLOv11-S, its mAP₅₀ is improved by 1.7% and 1.0%, respectively, and it also outperforms RT-DETRv2-L, which achieves 0.943. Although RT-DETRv2-L obtains a comparable mAP₅₀, its lower F1 score suggests weaker recall performance, whereas IADF-YOLO achieves a better precision–recall balance for small targets in complex scenes.

In terms of computational efficiency, IADF-YOLO requires only 5.8M parameters and 11.9 GFLOPs, corresponding to reductions of approximately 81.9% and 89.2%, respectively, compared with RT-DETRv1-L. In addition, it is more lightweight than YOLOv5-S, YOLOv6-3.0-S, and YOLOv8-S, while still achieving higher detection accuracy. IADF-YOLO runs at 61.1 FPS, outperforming YOLOv6-3.0-S and YOLOv8-S, thereby maintaining real-time performance together with high accuracy.

Furthermore, IADF-YOLO consistently achieves high and stable F1 scores on both datasets, reaching 0.92 on ISAD-6 and 0.89 on ISA-TD, both of which are higher than those of the competing methods. Overall, IADF-YOLO achieves a favorable trade-off among detection accuracy, computational efficiency, and real-time speed, further demonstrating its effectiveness and practical value for infrared small target detection under complex backgrounds and low signal-to-noise conditions.

3.4. Ablation experiment

Table 4 presents the ablation results for IADF-Net, CACSPFN, and TwoBackbone. It should be noted that, in this ablation study, *IADF-Net* and *TwoBackbone* are evaluated under two different settings. Specifically, the *IADF-Net* setting refers to using IADF-Net as an offline image enhancement module to pre-process the input images before they are fed into YOLOv12, without modifying the detector architecture or its parameters. In contrast, *TwoBackbone* refers to the dual-backbone training strategy, in which IADF-Net and YOLOv12 are jointly used during training. The results show that the baseline model achieves an mAP₅₀ of 0.941 and an mAP₅₀₋₉₅ of 0.616, with 9.2M parameters and 59.2 FPS. When IADF-Net is introduced as an offline enhancement module, the performance improves slightly to 0.942 in mAP₅₀ and 0.617 in mAP₅₀₋₉₅, while the inference speed remains unchanged. Enabling CACSPFN reduces the model size to 4.4M parameters and 10.4 GFLOPs, while increasing the FPS to 62.5. When TwoBackbone and CACSPFN are used together, the model achieves the best overall performance, reaching 5.8M parameters, 11.9 GFLOPs, an mAP₅₀ of 0.949, an mAP₅₀₋₉₅ of 0.624, and 61.1 FPS. These results indicate that the proposed design achieves a favorable balance between detection accuracy and computational efficiency.

Building upon the above analysis, Table 5 compares the effects of different image enhancement networks on infrared small target detection within a unified YOLOv12 detection framework. To ensure a fair comparison, all experiments use the same detection backbone and training settings, with only the parallel auxiliary enhancement branch being replaced. The results show that general-purpose enhancement models, including DeepLPP, Restormer, RetinexNet, MIRNet, and SNR-Net, can improve detection performance to some extent. However, these methods are primarily designed for generic image enhancement or restoration, rather than for weak-target preservation and degradation suppression

Table 5

Performance comparison of different image enhancement networks within the YOLOv12 framework.

Model	mAP ₅₀	mAP ₅₀₋₉₅	Params (M)	GFLOPs
DeepLPP (Moran et al., 2020) + YOLOv12 (Tian et al., 2025)	0.939	0.610	33.18	49.4
Restormer (Cai et al., 2023) + YOLOv12	0.938	0.608	45.86	77.9
RetinexNet (Wei et al., 2018) + YOLOv12	0.936	0.616	35.33	57.2
MIRNet (Kong et al., 2025) + YOLOv12	0.934	0.613	36.85	49.0
SNR-Net (Huang et al., 2023) + YOLOv12	0.941	0.611	29.46	43.2
IADF-Net + YOLOv12	0.945	0.620	11.90	25.6

Table 6

Performance of different FPN combinations with YOLOv12.

Model	mAP ₅₀	mAP ₅₀₋₉₅	Params (M)	GFLOPs
RepGFPN (Xu et al., 2022) + YOLOv12 (Tian et al., 2025)	0.927	0.606	12.25	29.7
CARAFE (Wang et al., 2019b) + YOLOv12	0.936	0.613	11.30	29.0
SDI (Ronneberger et al., 2015) + YOLOv12	0.930	0.610	18.84	28.0
BiFPN (Tan et al., 2020) + YOLOv12	0.935	0.612	10.24	28.8
CSFM (Zhao et al., 2024) + YOLOv12	0.923	0.601	7.26	23.2
CACSPFN + YOLOv12	0.942	0.619	4.40	10.4

in infrared small target detection. In contrast, under the same setting, IADF-Net achieves superior overall performance while also maintaining lower parameter count and computational cost. This indicates that IADF-Net is not simply a substitute for existing illumination enhancement models, but rather a detection-oriented degradation filtering module that is better aligned with the requirements of infrared small target detection. This comparative study further demonstrates that IADF-Net differs fundamentally from general illumination enhancement models such as RetinexNet and Retinexformer in both task objective and design paradigm. Its primary role is not to improve visual quality, but to provide more discriminative feature representations for infrared small target detection.

In addition, Table 6 reports the experimental results of different FPN variants integrated into YOLOv12. Among all the compared methods, the CACSPFN + YOLOv12 configuration achieves the best overall performance, yielding an mAP₅₀ of 0.942 and an mAP₅₀₋₉₅ of 0.619. Notably, these improvements are achieved with only 4.4M parameters and 10.4 GFLOPs, indicating that the proposed CACSPFN effectively enhances detection accuracy while maintaining low computational cost. These results further confirm the effectiveness and efficiency of CACSPFN for infrared small target detection.

3.5. Visualization and analysis

To better understand the training dynamics and representation quality of IADF-YOLO, we present a visual analysis of its convergence behavior, feature responses, and qualitative detection results by jointly examining the loss curves (Fig. 6), response heatmaps (Fig. 7), and detection examples (Fig. 8), without repeating the architectural details. On the ISAD-6 dataset, IADF-YOLO consistently outperforms YOLOv12 and Gold-YOLO in terms of convergence speed, training stability, and final loss. Although lightweight detectors typically exhibit slower optimization, the training-time synergy between IADF-Net and CACSPFN enables faster and more stable early convergence, while maintaining lower box regression and classification losses throughout training, indicating improved localization accuracy and feature discriminability. The feature-response heatmaps further show that IADF-YOLO produces compact and well-localized activations around true targets while effectively suppressing background interference. In contrast, YOLOv8, YOLOv11, and RT-DETRv2 exhibit scattered or spurious responses in cluttered terrain and long-range scenes, reflecting greater sensitivity to background noise. The qualitative results further support these observations: IADF-YOLO delivers more stable confidence scores and more accurate

localization under both near-range and long-range conditions, whereas the competing methods suffer from missed detections, confidence degradation, or localization drift. Overall, the visual evidence demonstrates the superior target-background separability and robustness of IADF-YOLO in challenging infrared scenarios.

4. Conclusion

From an application perspective, the proposed IADF-YOLO demonstrates strong robustness and stability for infrared small target detection in complex scenarios. By introducing illumination-aware degradation constraints during training and enhancing the exploitation of discriminative cues in multi-scale feature representations, the model can effectively distinguish true targets from background interference under challenging conditions, such as low signal-to-noise ratios, complex backgrounds, and long-range observation. Moreover, because only the conventional YOLO backbone is retained during inference, IADF-YOLO improves feature quality while preserving a lightweight inference architecture, thereby satisfying the requirements of real-time infrared sensing systems in terms of computational efficiency and deployment flexibility. These characteristics make the proposed method well suited for practical applications such as intelligent surveillance, remote early warning, and unmanned system perception.

Nevertheless, IADF-YOLO still has certain limitations under extreme conditions. When infrared images contain large-scale non-uniform radiation, severe background fluctuations, or highly similar thermal patterns between targets and background, the effectiveness of degradation suppression and feature enhancement may be reduced. In addition, under severe occlusion, dense target distributions, or extremely low-resolution inputs with highly imbalanced illumination, there is still room for further improvement in detection stability. Future work will focus on incorporating more diverse infrared data, together with transfer learning and domain adaptation strategies, as well as integrating temporal modeling to further enhance robustness and reduce false alarms in dynamic and complex environments.

CRedit authorship contribution statement

Shao-Qi Sun: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Li-Juan Liu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Hamid Reza Karimi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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